

Supplementary Information

Real-time Monitoring and Hydrodynamic Scaling of Shear Exfoliated Graphene

Jason Stafford,^{1, 2} * Nwachukwu Uzo,¹ Usmaan Farooq,¹ Silvia Favero,¹ Si Wang,¹ Hsueh-Hung Chen,¹ Anouk L'Hermitte,¹ Camille Petit,¹ and Omar K. Matar¹.

¹ Department of Chemical Engineering, Imperial College London, SW7 2AZ, United Kingdom.

² School of Engineering, College of Engineering & Physical Sciences, University of Birmingham, B15 2TT, United Kingdom.

* j.stafford@bham.ac.uk

Supplementary information included with this article are:

Sections S1 to S7 below.

Movie S1. Shear exfoliation of graphite.

Movie S2. Taylor vortex flow and graphite precursor motion.

Movie S3. Direct numerical simulations and experimental spinning disc flows.

Movie S4. Real-time monitoring of liquid exfoliation.

Section S1. Liquid exfoliation device details

The following table provides geometric details for both devices. For the Taylor-Couette device, r_i is the radius of the inner rotating cylinder, r_o is the radius of the outer stationary cylinder, and h is the cylinder height. In the case of the spinning disc device, r_i is the radius at which fluid from the nozzle exits onto the disc, r_o is the radius of the disc, and h is the nozzle exit height.

Table S1. Summary of exfoliation device dimensions.

Exfoliation device	r_i (mm)	r_o (mm)	h (mm)
Taylor-Couette	50.5	52.5	100
Spinning disc	21	100	2

Section S2. Additional materials characterization detail

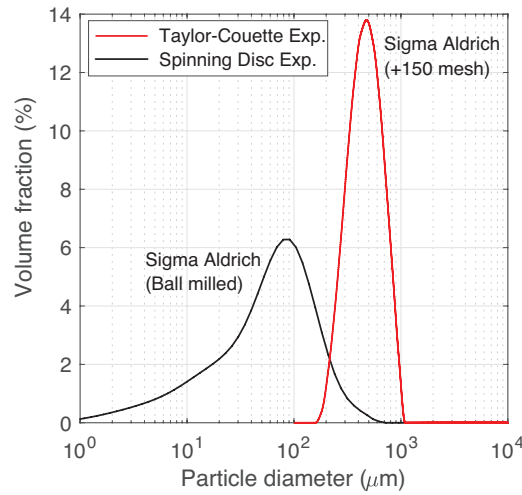


Figure S1. Graphite input size distribution.

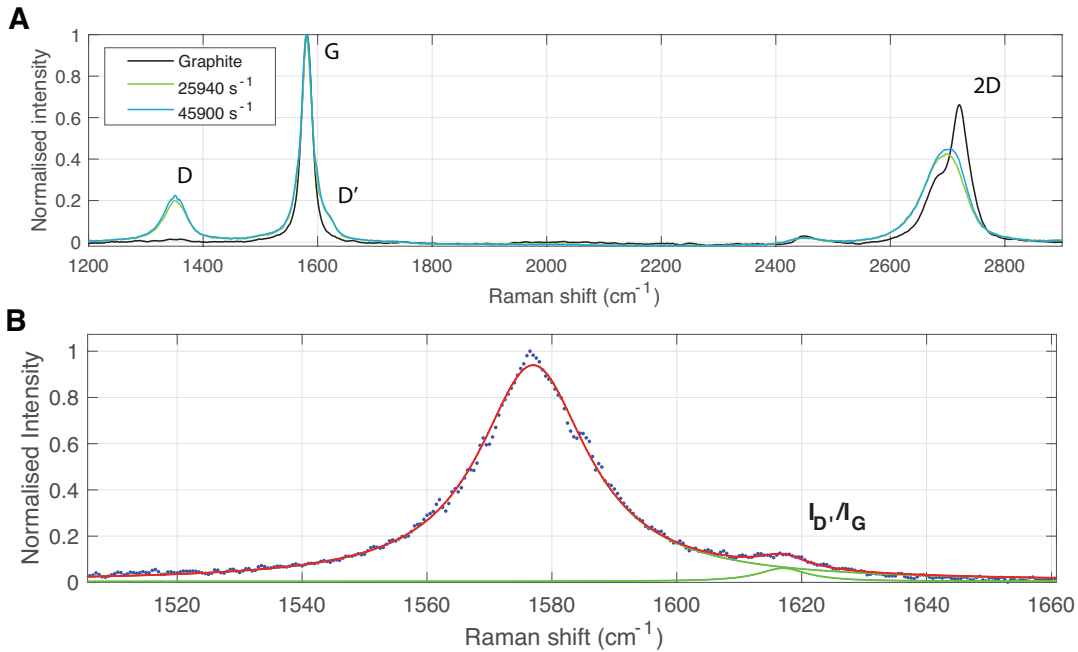


Figure S2. Raman spectra and analysis for graphite and graphene. (A) Complete Raman spectra for graphite and graphene. **(B)** I_G and $I_{D'}$ including peak-fitting. Peaks were fitted using Lorentzians. In this example ($C_i = 10 \text{ g L}^{-1}$, $\langle \dot{\gamma}_w \rangle = 45900 \text{ s}^{-1}$), the error and R^2 are 1.327% and 0.996. Analyses derived from tests run on the Taylor-Couette device.

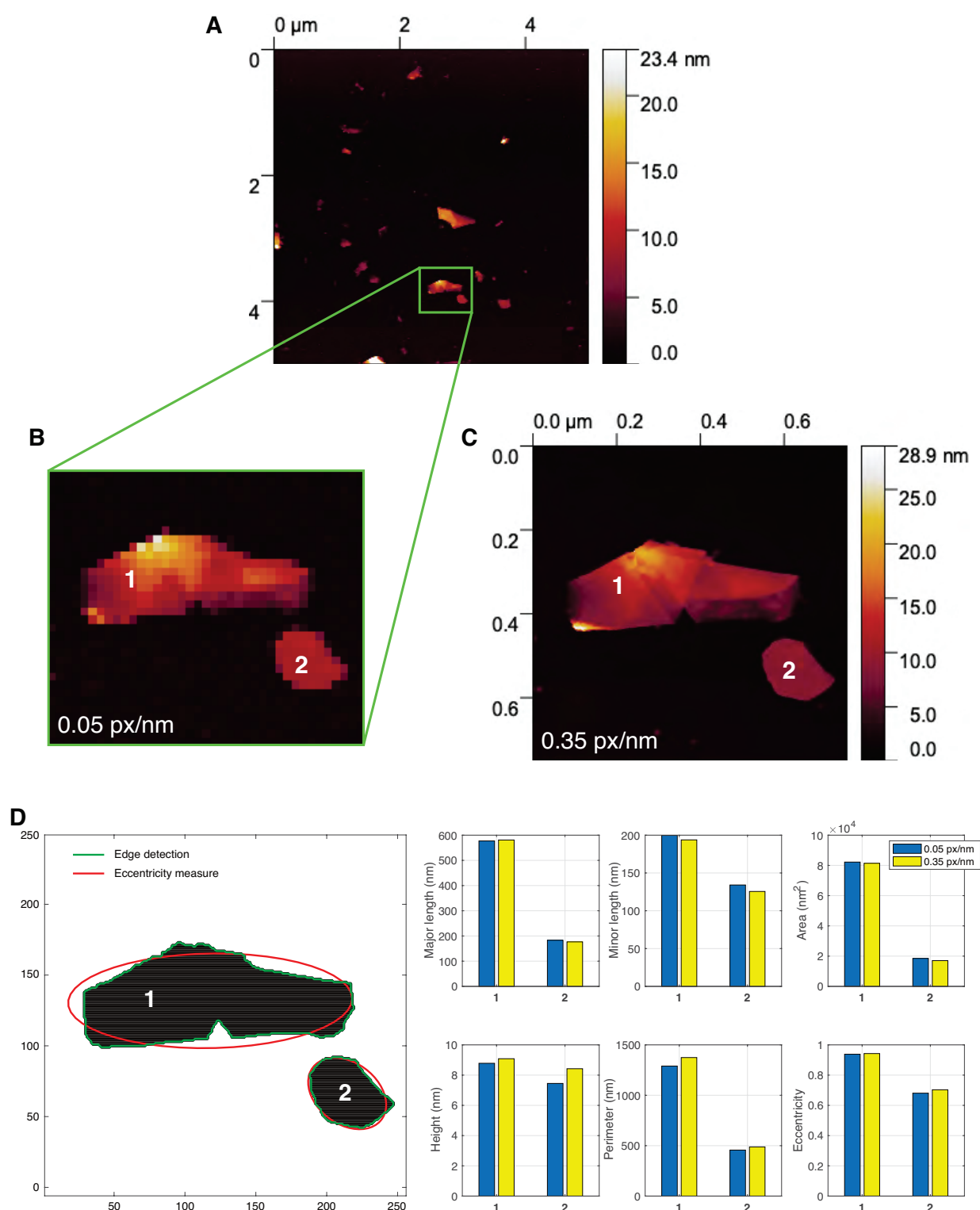


Figure S3. Assessing measurement sensitivity to AFM resolution. (A) AFM measurement of graphene nanosheets with x-y resolution of 0.05 px/nm. (B) A zoomed view on two nanosheets. (C) AFM measurement of the same two graphene nanosheets with x-y resolution of 0.35 px/nm. (D) Edge detection and eccentricity measurements on sample nanosheets (left). Comparisons between nanosheet parameters of major length, minor length, area, height, perimeter, and eccentricity (right). Analyses derived from tests run on the Taylor-Couette device.

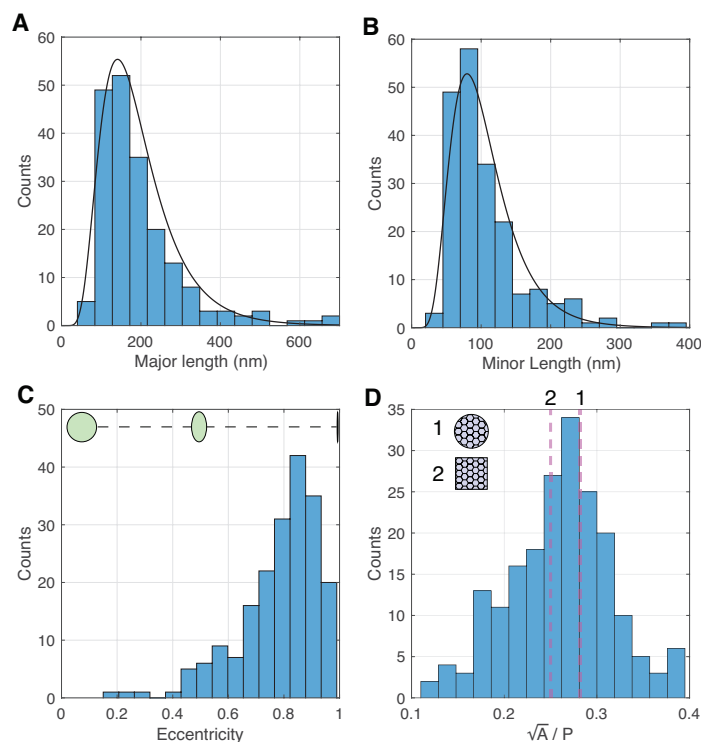


Figure S4. Nanosheet length and shape distributions. (A) Major and (B) minor lengths. (C) Eccentricity and (D) $\sqrt{\text{Area}}/\text{Perimeter}$ parameters. ($C_i = 10 \text{ g L}^{-1}$, $\langle \dot{\gamma}_w \rangle = 32120 \text{ s}^{-1}$). Analyses derived from tests run on the Taylor-Couette device.

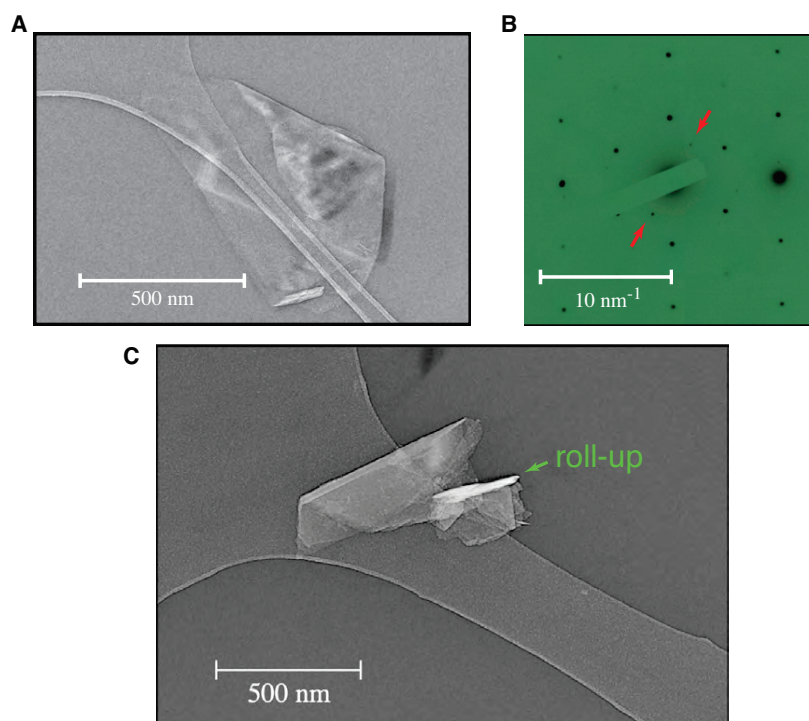
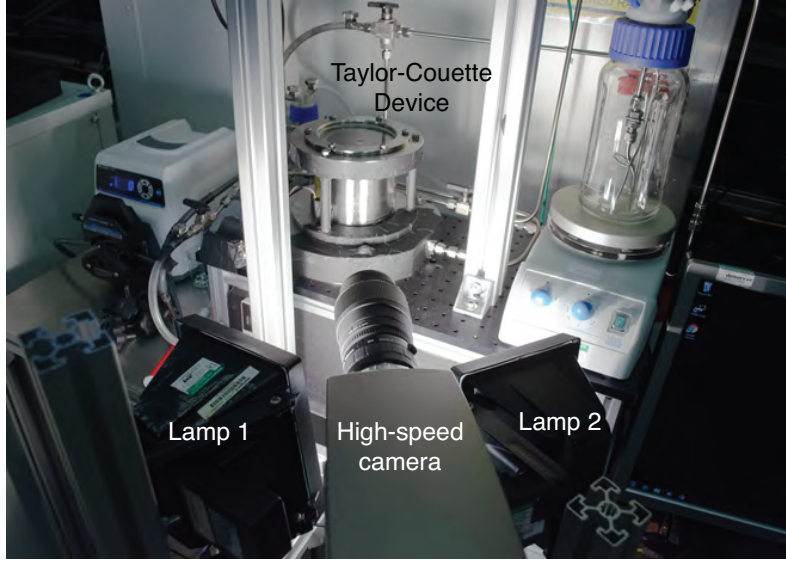


Figure S5. TEM images of nanosheet roll-up. (A) A folded and partially rolled graphene sheet observed with TEM and (B) the diffraction pattern. The inner peaks indicate rolling with an interlayer spacing of 0.34 nm, similar to multi-walled carbon nanotubes (MWCNTs). Analyses derived from tests run on Taylor-Couette device ($C_i = 10 \text{ g L}^{-1}$, $\langle \dot{\gamma}_w \rangle = 32120 \text{ s}^{-1}$). (C) TEM image of nanosheet roll-up in the spinning disc exfoliation device ($C_i = 5 \text{ g L}^{-1}$, $\langle \dot{\gamma}_w \rangle = 32455 \text{ s}^{-1}$).

Section S3. Graphite tracking experiment

A



B

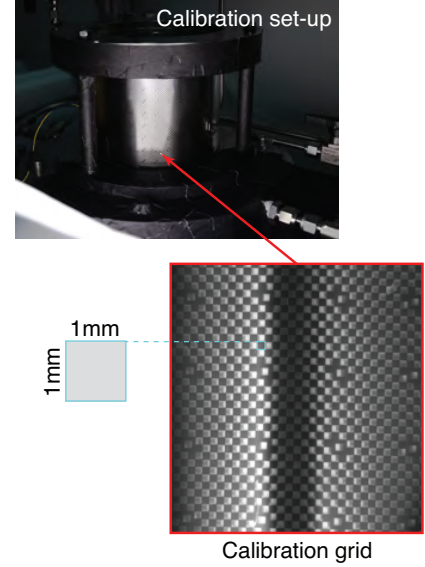


Figure S6. Tracking of graphite motion over short timescales. (A) Experimental configuration including high speed camera and two halogen lamps as light sources. (B) Spatial calibration set-up for particle tracking and image velocimetry.

Section S4. Governing equations for DNS and LES

Conservation equations for mass and momentum:

$$\nabla \cdot \mathbf{u} = 0 \quad (1)$$

$$\rho \left[\frac{\partial \mathbf{u}}{\partial t} + \nabla \cdot (\mathbf{u}\mathbf{u}) \right] = -\nabla p + \nabla \cdot [2\mu \mathbf{S}] + \rho \mathbf{g} + \mathbf{F} \quad (2)$$

in which $\mathbf{u}$ is the velocity vector, p denotes pressure, $\mathbf{g}$ the gravitational acceleration, ρ and μ represent the density and viscosity of the fluid, respectively, and t is time. The force $\mathbf{F}$ for the case of the spinning disc represents the surface force, appended to the Navier-Stokes equations, and given by $\mathbf{F} = \sigma \kappa \mathbf{n}$, wherein σ is the surface tension, κ the interfacial curvature, and $\mathbf{n}$ the outward pointing unit normal to the interface. The rate of deformation tensor, $\mathbf{S}$, is given by:

$$\mathbf{S} = \frac{1}{2} [\nabla \mathbf{u} + \nabla \mathbf{u}^T] \quad (3)$$

Section S4.1. Large-Eddy simulations for Taylor-Couette exfoliation

Direct numerical simulations of the governing equations above for turbulent flows with Reynolds numbers examined ($Re \sim 10^4$) can have considerable computational expense (time and resource). Instead, we performed Large-Eddy simulations (LES) to obtain solutions that capture turbulent motion down to a point, beyond which a sub-grid model is implemented for smaller scales. The filtered equations governing the resolved flow are:

$$\nabla \cdot \bar{\mathbf{u}} = 0 \quad (4)$$

$$\frac{\partial \bar{\mathbf{u}}}{\partial t} + \nabla \cdot (\bar{\mathbf{u}}\bar{\mathbf{u}}) = -\frac{1}{\rho} \nabla \bar{p} + \nabla \cdot [2\nu_e \bar{\mathbf{S}}] + \rho \mathbf{g} \quad (5)$$

where $\nu_e = \nu + \nu_t$ is the effective viscosity consisting of the molecular and eddy viscosities, respectively. In the LES approach, scales smaller than the grid size are not resolved but accounted for through a sub-grid-scale tensor $\mathbf{T}_{sgs}$ given by:

$$\mathbf{T}_{sgs} = \overline{\mathbf{u}\mathbf{u}} - \overline{\mathbf{u}}\overline{\mathbf{u}} \quad (6)$$

where the overbar denotes an appropriately chosen low-pass filter, and incompressibility is assumed. We applied an implicit top-hat filter, where $\Delta = V_{cell}^{1/3}$, and V_{cell} is the cell volume in the numerical grid. Most sub-grid-scale cell models are based on an eddy-viscosity assumption to model the sub-grid-scale tensor:

$$\mathbf{T}_{sgs} - \frac{1}{3} \text{tr}(\mathbf{T}_{sgs})\mathbf{I} = 2\nu_t \bar{\mathbf{S}} \quad (7)$$

where $\mathbf{I}$ is the identity tensor, and $\bar{\mathbf{S}}$ is the deformation tensor of the resolved field:

$$\bar{\mathbf{S}} = \frac{1}{2} [\nabla \bar{\mathbf{u}} + \nabla \bar{\mathbf{u}}^T] \quad (8)$$

We implemented a Wall Adapting Local Eddy-Viscosity (WALE) approach which is invariant to coordinate translation/rotation, is a function of both strain and rotation rates, and naturally tends to zero at the wall which removes the need for a damping function or dynamic procedure to reproduce the effect of the no-slip condition (46). The sub-grid-scale eddy-viscosity for the WALE approach is:

$$\nu_t = (C_w \Delta)^2 \frac{(\mathcal{S}_{ij}^d \mathcal{S}_{ij}^d)^{3/2}}{(\bar{\mathcal{S}}_{ij} \bar{\mathcal{S}}_{ij})^{5/2} + (\mathcal{S}_{ij}^d \mathcal{S}_{ij}^d)^{5/4}} \quad (10)$$

where C_w is a model constant, Δ is the sub-grid characteristic length (grid size), and $\mathcal{S}_{ij}^d$ is the traceless symmetric part of the velocity gradient tensor.

Section S4.2. Validation of Taylor-Couette simulations

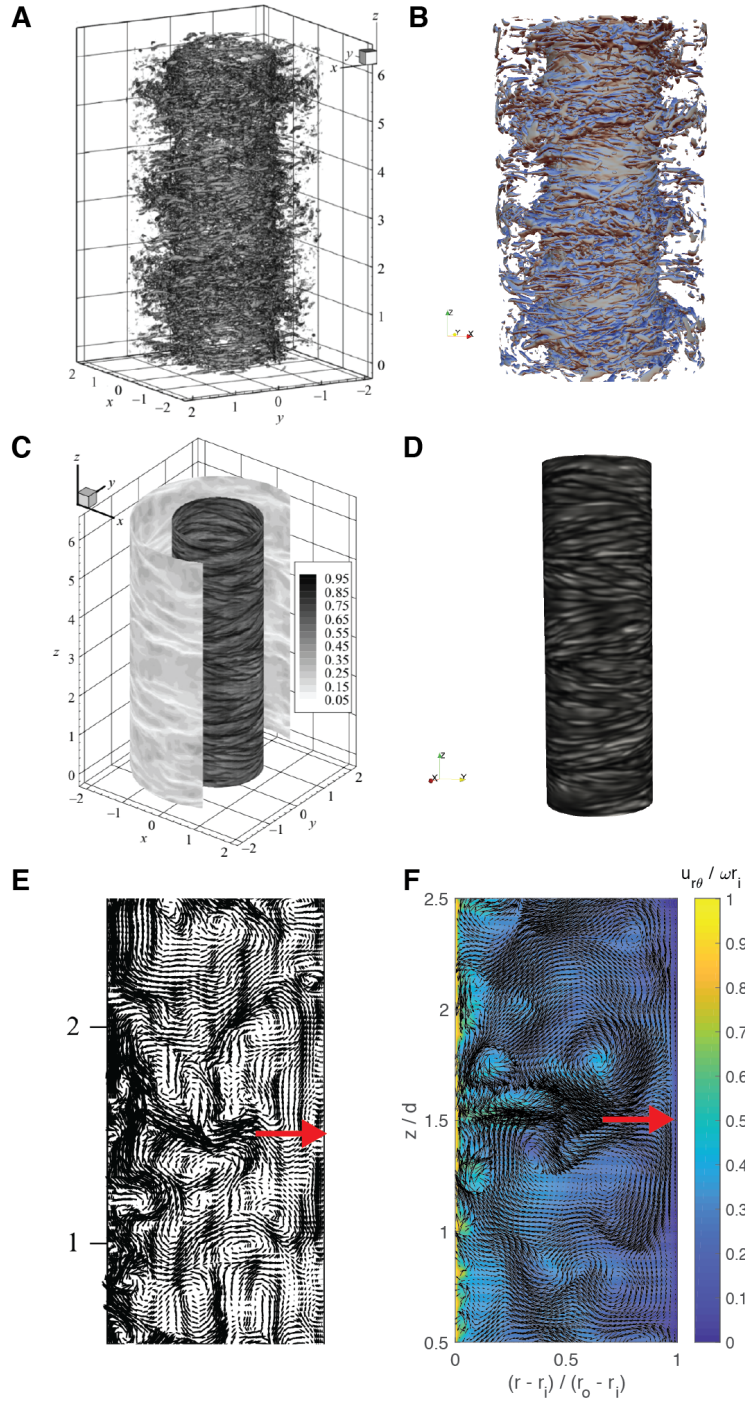


Figure S7: Comparisons between Direct Numerical Simulations (48) and Large-Eddy Simulations (right). (A),(B) Instantaneous isocontours of λ_2 criterion, illustrating vortices in this flow. In this identification method (51), the pressure minimum in a plane perpendicular to a vortex axis is found when the second of three eigenvalues ($\lambda_1 \geq \lambda_2 \geq \lambda_3$) of the symmetric tensor $\mathbf{S}^2 + \mathbf{\Omega}^2$ is negative, with $\mathbf{S}$ and $\mathbf{\Omega}$ the symmetric and antisymmetric parts of the velocity gradient tensor $\nabla \mathbf{u}$. (C),(D) wall shear stress distribution on the inner rotating cylinder. (E),(F) Instantaneous velocity field. Color scale represents the velocity magnitude, $u_{r\theta}$, calculated from (u_r, u_θ) velocity components and normalised by the inner cylinder speed, ωr_i . Velocity vectors in the r - z plane are represented by (u_r, u_z) . Here, r , z and θ represent the radial, axial and azimuthal coordinates, respectively, ω the rotational speed of the inner cylinder, located at $r = r_i$, and r_o is the radius of the outer cylinder.

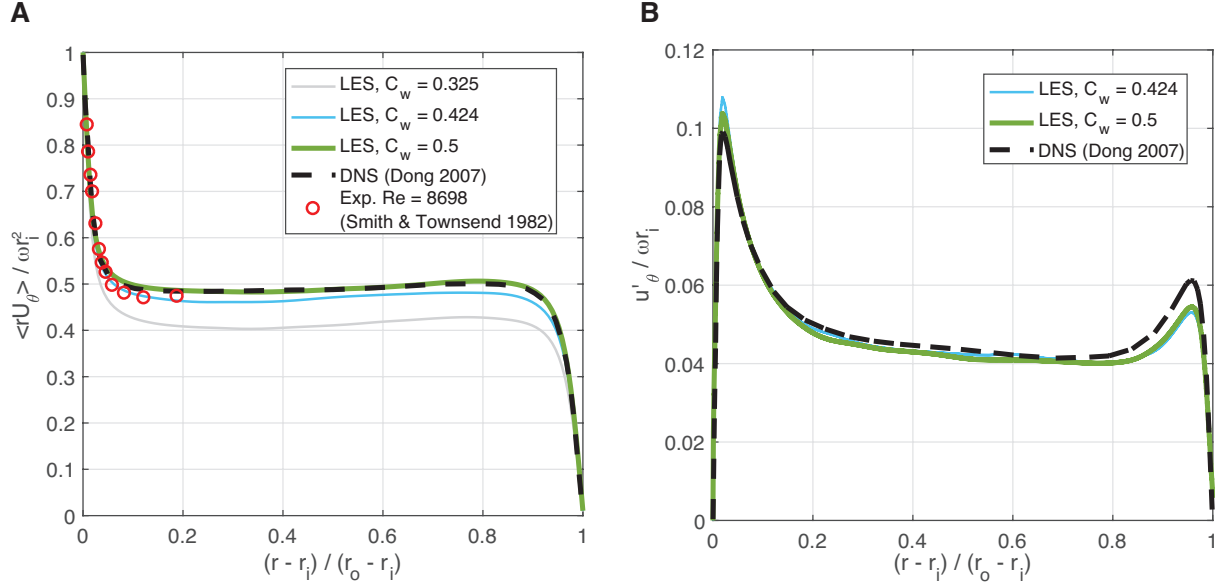


Figure S8: Assessment and validation of the LES subgrid-scale model. (A) Angular momentum profiles for different model constant C_w that appears in Eq. (10) compared with DNS and experimental data from the literature (48,49). (B) Root-mean-square azimuthal velocity fluctuation profiles for different model constants compared with DNS (48).

Section S4.3. Direct numerical simulations for spinning disc exfoliation

Direct numerical simulations were used to solve directly the governing equations given by equations (1) and (2), of the thin film flows over the spinning disc. The liquid-air interface is captured using a geometric Volume-of-Fluid (VoF) approach (47), within the framework of which $\mathbf{F} = \sigma \kappa \nabla \alpha$, where α is the phase fraction. Intrinsic properties of the different fluids (ρ, μ) are obtained as follows:

$$\rho = \alpha \rho_{\text{nmp}} + (1 - \alpha) \rho_{\text{air}} \quad (11)$$

$$\mu = \alpha \mu_{\text{nmp}} + (1 - \alpha) \mu_{\text{air}} \quad (12)$$

and α evolves according to:

$$\frac{\partial \alpha}{\partial t} + \nabla \cdot (\alpha \mathbf{u}) = 0 \quad (13)$$

This equation is not solved directly in the VoF method implemented. The interface location is identified and it is then advected in a geometric manner (47).

Section S4.4. Validation of spinning disc simulations

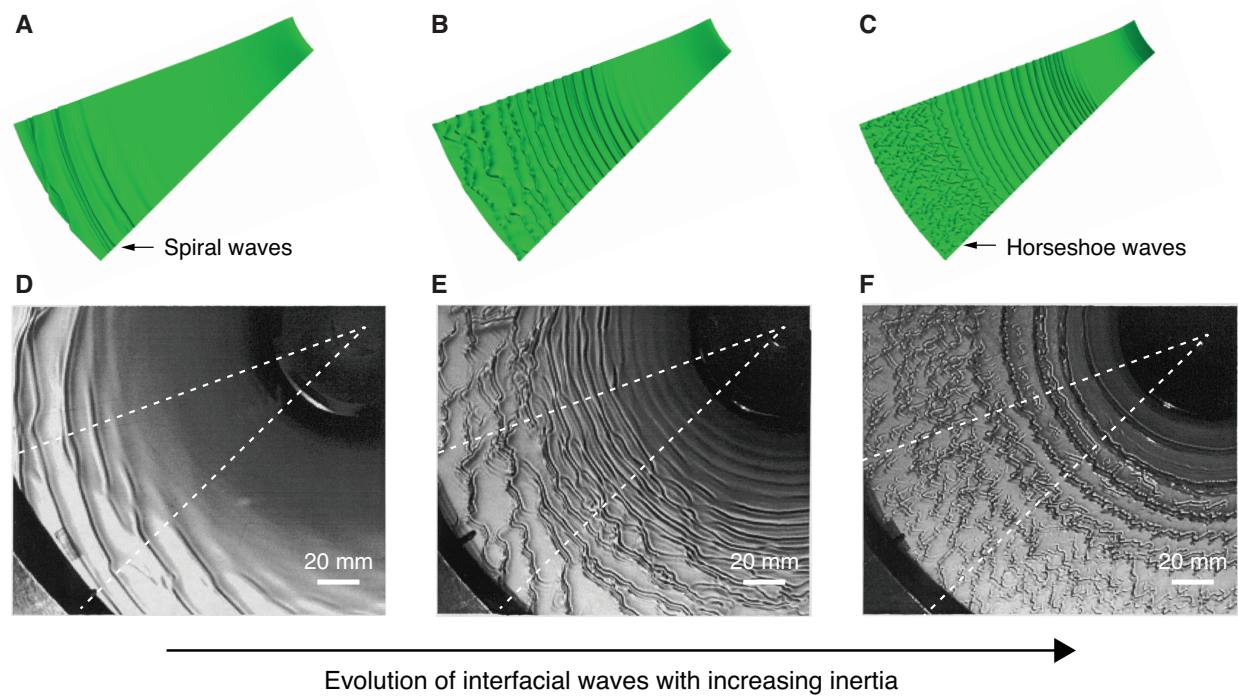


Figure S9: Comparisons between Direct Numerical Simulations (top) and experiments (bottom). (A to C) Instantaneous isocontours of $\alpha = 0.5$ illustrating the interface topology. (D to F) Photographs of a thin film flowing over a rapidly rotating disc (50). This validation example has a disc radius of 180 mm, flow rate of 1140 mL/min, and rotational speeds (A),(D) 10.47 rad/s, (B),(E) 20.94 rad/s, and (C),(F) 62.83 rad/s, respectively.

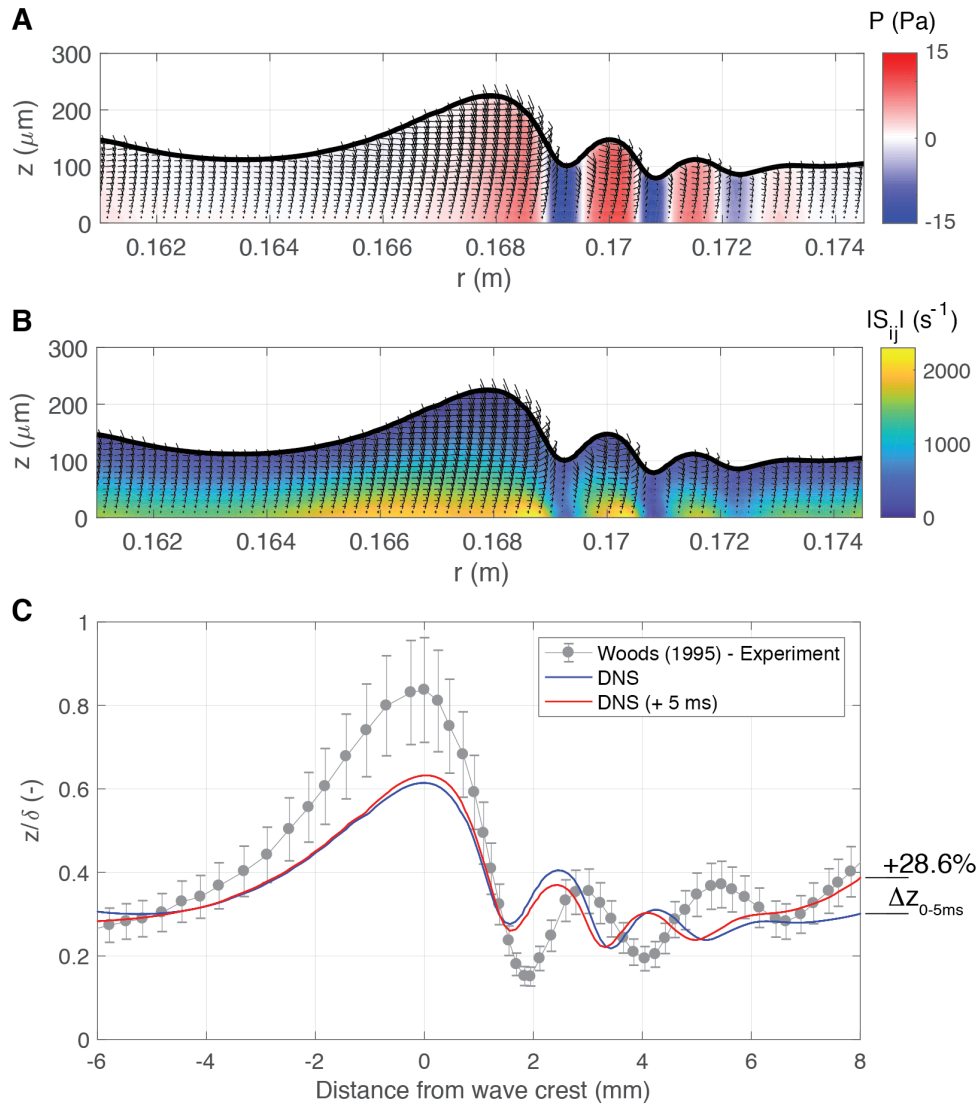


Figure S10: Assessment and validation of DNS. (A) Pressure distribution in the liquid film and under a wave. Velocity vectors in the r - z plane are represented by (u_r, u_z) . Here, r and z represent the radial and axial coordinates. (B) Magnitude of the strain rate distribution. (C) A comparison between predicted and experimental (50) film height for a flow rate of 1140 mL/min and a rotational speed of 10.47 rad/s. δ represents the waveless film height obtained from Nusselt theory.

Section S5. Local strain rate scaling

Section S5.1. Taylor-Couette

The following figures present the local strain rate distributions in the Taylor-Couette exfoliation device (Fig. 1, Fig. 3A,B, Fig. 5, Fig. 6).

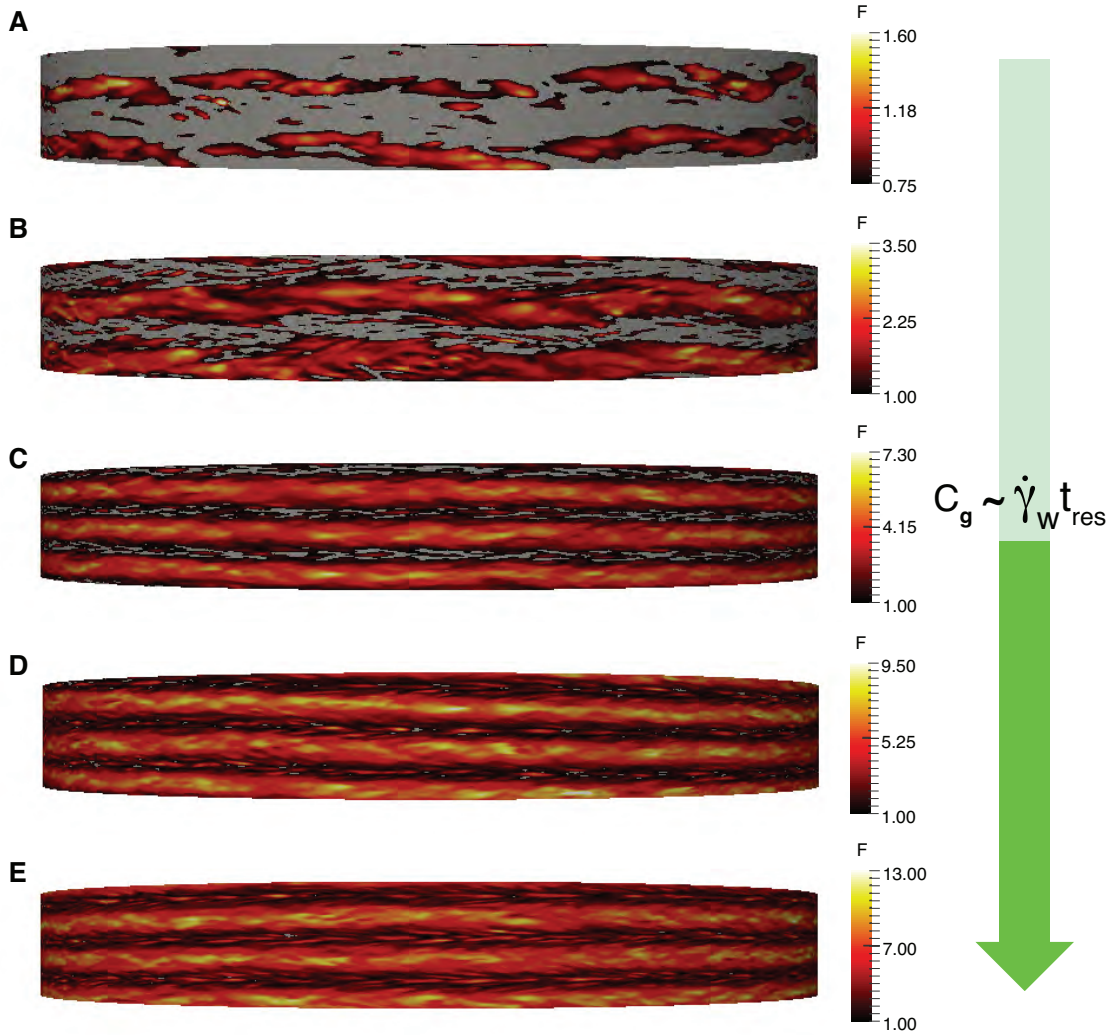


Figure S11. Local strain rate for Taylor-Couette flow. (A) 50.5 rad/s, $F > 0.75$. (B) 84 rad/s, $F > 1$. (C) 118 rad/s, $F > 1$. (D) 151 rad/s, $F > 1$. (E) 185 rad/s, $F > 1$. The strain rate data are represented by $F = \dot{\gamma}_w / \dot{\gamma}_{crit}$, where $\dot{\gamma}_{crit} = 10^4 \text{ s}^{-1}$.

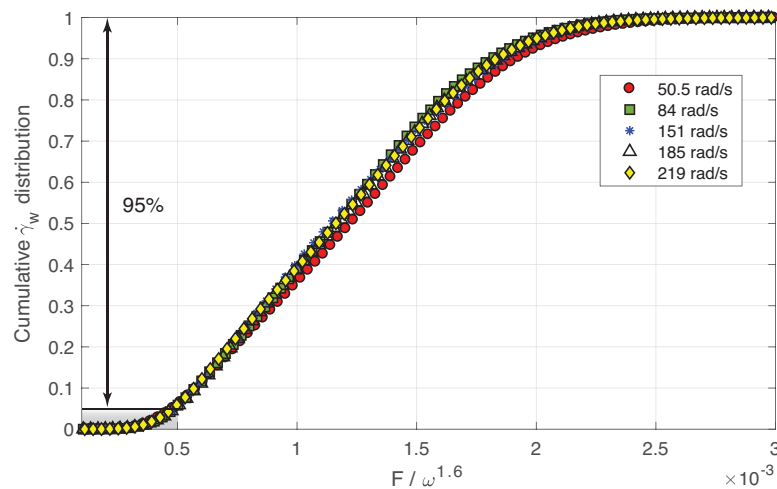


Figure S12. Strain rate histogram for Taylor-Couette flow. Cumulative distributions normalized by the global momentum transport scaling, $\omega^{1.6}$.

Section S5.2. Spinning disc

The following figures present the local strain rate distributions in the spinning disc exfoliation device (Fig. 2, Fig. 3C,D).

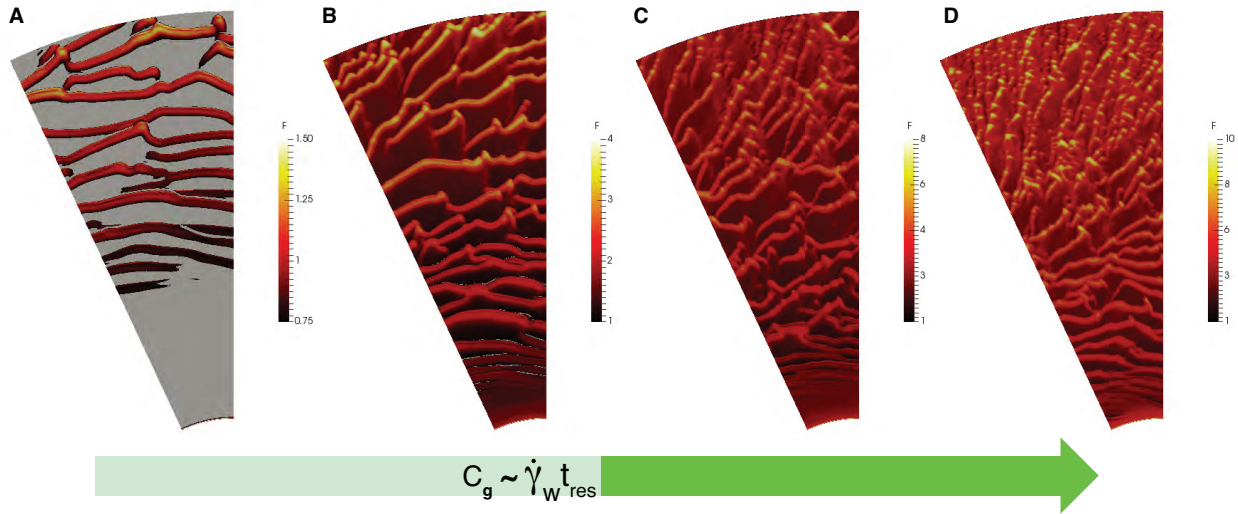


Figure S13. Local strain rate for Spinning disc flow. (A) 52 rad/s, $F > 0.75$. (B) 105 rad/s, $F > 1$. (C) 157 rad/s, $F > 1$. (D) 209 rad/s, $F > 1$. The strain rate data are represented by $F = \dot{\gamma}_w / \dot{\gamma}_{crit}$, where $\dot{\gamma}_{crit} = 10^4 \text{ s}^{-1}$. Flow rate of 360 mL/min.

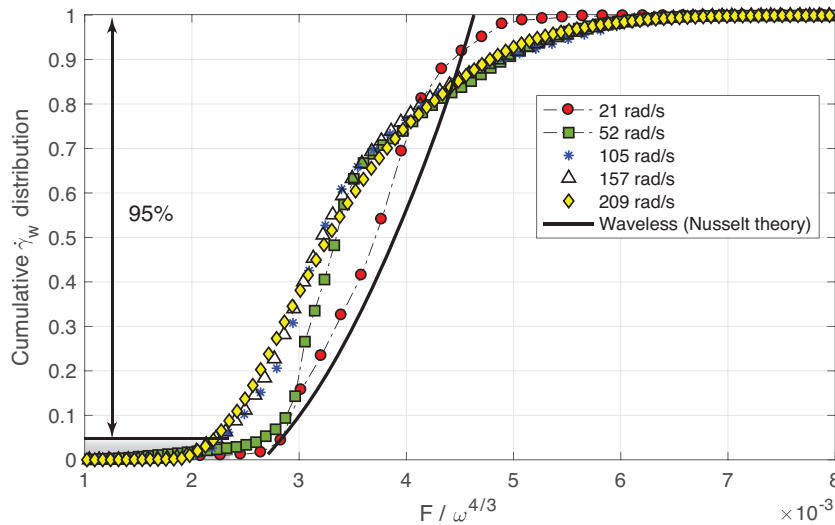


Figure S14. Strain rate histograms for spinning disc flow. Cumulative distributions normalized by the global momentum transport scaling, $\omega^{4/3}$. There is a transition in wave regimes between 21 rad/s and 105 rad/s, and 21 rad/s data closely aligns with Nusselt theory for waveless thin films. The differences with waveless thin film theory in the lowest speed case are due to 1-2 waves emerging at the disc periphery in an otherwise smooth film. Flow rate of 360 mL/min.

Section S5.3. Microfluidization

In addition to the two test cases studied in this work, we examined independently published data on microfluidization. This is a shear-driven liquid exfoliation technique which has received particular attention for large-scale production. This technique can produce strain rates far exceeding the critical exfoliation criteria and typically $\dot{\gamma}_w > 10^4 \text{ s}^{-1}$ (17-19). Arao *et al.* (17) investigated graphene production performance of microfluidization using an aqueous solution

(water/TritonX-100). The authors varied the applied pressure and measured the resultant graphene concentration after centrifugation (Fig. S15A).

A force balance for the laminar exfoliation zone shows:

$$\langle \dot{\gamma}_w \rangle = \frac{A \Delta P}{\mu p L} \quad (14)$$

where A , p , and L are the channel cross-sectional area, perimeter and length, respectively. The applied pressure difference scales linearly with wall shear stress, and for solvents which behave as Newtonian fluids, $\Delta P \sim \dot{\gamma}_w$. In Fig. S15B, the graphene concentration data is rescaled in the form of $C_g / \dot{\gamma}_w t_{res}$. The data collapse to a constant $\approx 2 \times 10^{-5}$, supporting the present observations that production output scales $C_g \sim \dot{\gamma}_w t_{res}$ when above criticality.

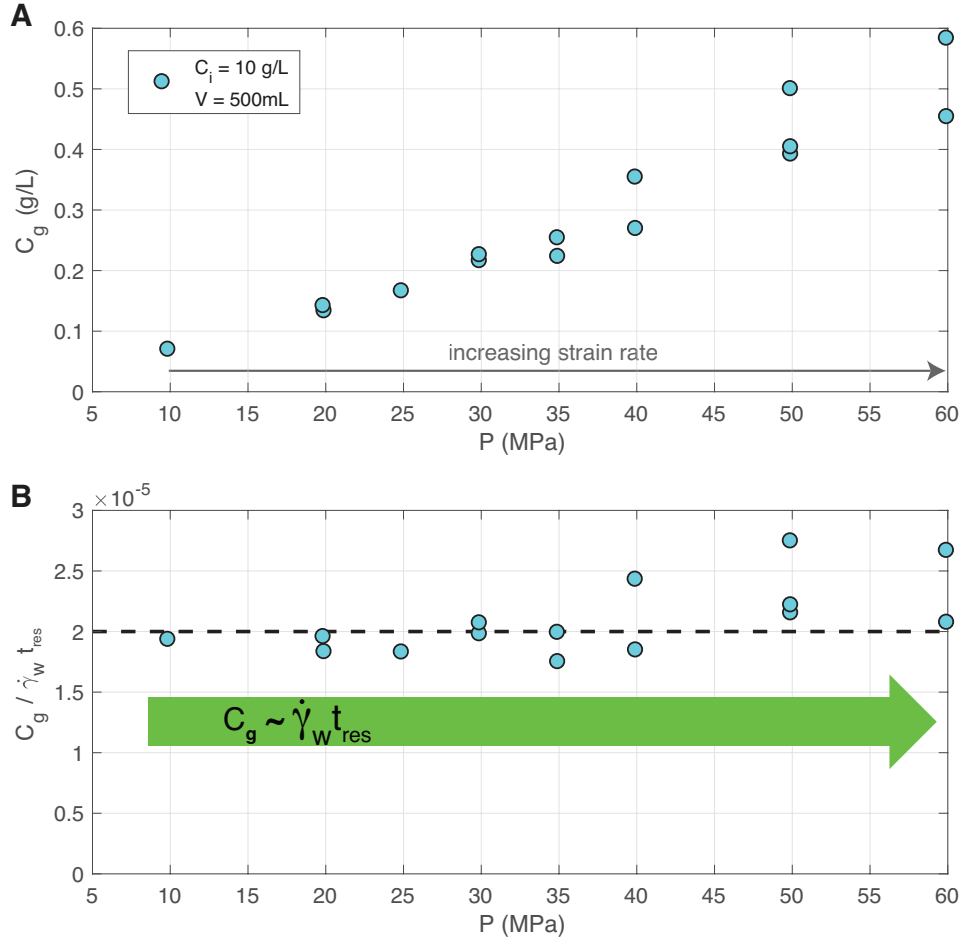


Figure S15. Rescaling of microfluidization exfoliation. (A) Graphene concentration for a range of applied pressures, replotted from Arao *et al.* (17). (B) Concentration data rescaled with strain rate, $C_g \sim \dot{\gamma}_w t_{res}$.

Section S6. Saturation time

After passing through the Taylor-Couette exfoliation zone (Z1), the precursor material flows into a second zone (Z2) before leaving the device (Fig. 1A). This additional zone allowed us to create different concentration profiles, that would subsequently be used to assess the *in situ* production monitoring technique (Figs. 5 and 6). Here, the precursor was forced to deposit on the surface of the inner cylinder over long processing times, and depending on rotational speed (Fig. 1D). This resulted in saturation of graphene production at different times (t_{sat}).

We can examine the scaling of t_{sat} by analyzing the dominant forces on a precursor particle within this region. First, we look at the forces acting on a particle of mass m , and density ρ_p , in a

fluid of uniform density ρ and kinematic viscosity ν . In a rotating reference frame, this includes gravitational, Coriolis and centrifugal body forces, forces related to hydrostatic and centrifugal pressure gradients within the solvent, and a hydrodynamic force ($\mathbf{F}_{\text{hyd}}$) related to the particle motion with respect to the fluid. A force balance on a particle can be described as:

$$m\mathbf{a} = \mathbf{F}_{\text{hyd}} - \Delta\rho V_p \mathbf{g} - 2m\boldsymbol{\omega} \times \mathbf{u} + \Delta\rho V_p \omega^2 r \hat{\mathbf{r}} \quad (15)$$

where m is the particle mass, $\Delta\rho = \rho_p - \rho$, V_p is the particle volume, $\mathbf{g}$ is gravity, $\mathbf{u}$ is the translating particle velocity, and $\mathbf{a}$ is particle acceleration. A scale analysis suggests the centrifugal force is a dominant term when $\omega \sim 10^2$ rad/s, and acceleration scales as $\mathbf{a} \sim \omega^2 r$. Therefore, the time it takes for a graphite particle to sediment should scale with rotational speed, $t_{\text{sat}} \sim \omega^{-1}$. Indeed, this is observed in Fig. 1E, as concentration data collapse to a constant $N_o \approx 10^5$, confirming the anticipated scaling of saturation time in the particle deposition region Z2.

Section S7. Multispectral transmission-reflectance spectroscopy

The following figures are supplementary information for the *in situ* exfoliation monitoring technique (Figs. 5 and 6).

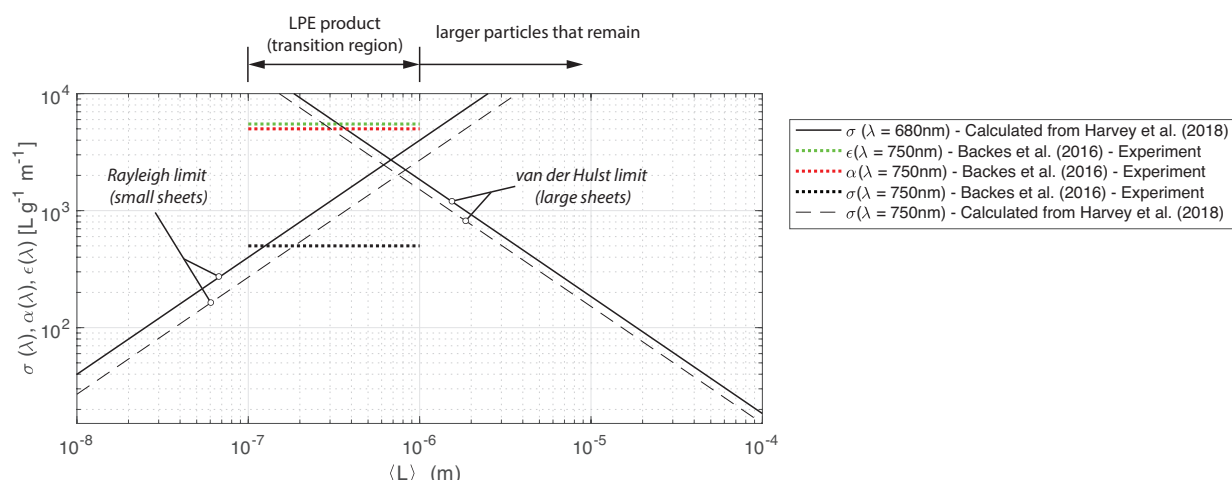


Figure S16. Relationship between particle size and scattering coefficient. Rayleigh and van der Hulst scattering limits for a range of average particle lengths (10 nm to 100 microns) and using the transmission-reflectance measurement wavelength (680 nm). For this calculation, we assumed the refractive indices of graphene and NMP ($n = 2.6$, $n = 1.47$), and solvent density (1000 kg m^{-3}). Also included are experimental data for $\lambda = 750 \text{ nm}$ from Backes et al. (11) who previously decomposed the extinction (ϵ), absorption (α) and scattering (σ) coefficients for LPE graphene. While there was some variation in the original experimental data, the coefficients were shown to be approximately constant for average lengths spanning 100 nm to 1 micron. Absorbance is shown to dominate measured extinction spectra, where $\epsilon(\lambda) = \alpha(\lambda) + \sigma(\lambda)$. The constant scattering behavior with particle size may occur as most LPE product is in this transitional region between the two scattering limits.

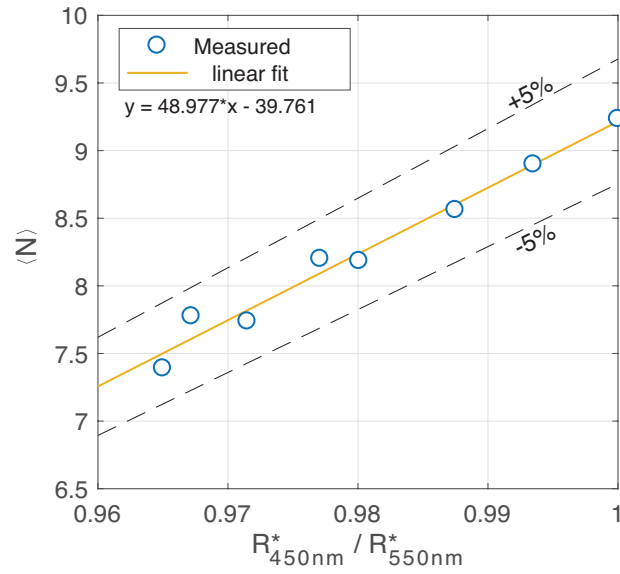


Figure S17. Relationship between number of atomic layers and *in situ* multispectral reflectance measurements at 450 nm and 550 nm (see Fig. 5B also). Reflectance has been normalized with the baseline reflectance for $\langle N \rangle \approx 9$ to account for the differences in spectral power at the two different wavelengths (e.g. $R_{450\text{nm}}^* = R_{450\text{nm}} / [R_{450\text{nm}} @ \langle N \rangle = 9]$).